Resizing a dynamically allocated array.

Idea: find array a larger new home and move all elements.

1. Allocate larger array
2. copy elements to new array
3. deallocate old array
4. redirect A to point to new array.

Linked Lists

Motivation: to insert a new element near the beginning of a vector or array is expensive (proportional to the # elements)

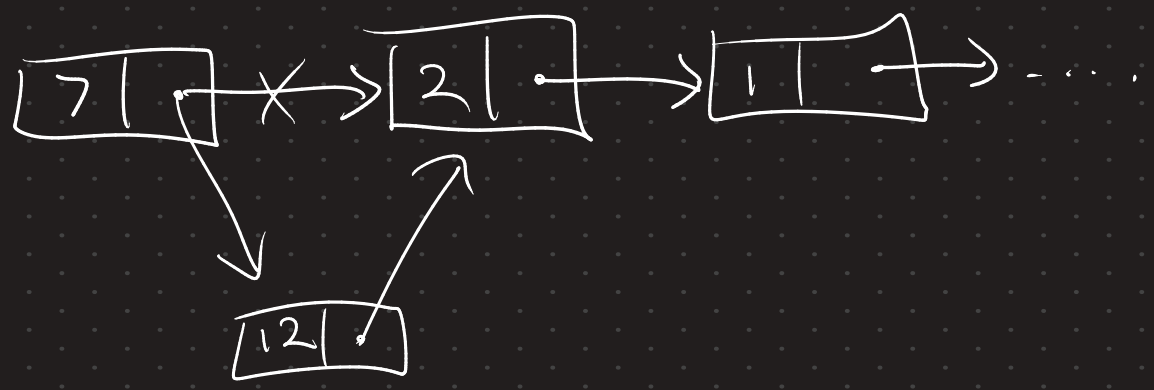
Idea: have each element tell you where

to find the next element.

(with a vector, there is no choice:

&V[i+1] == (&V[i]) + 1)

Pictures:



want to associate a pointer to each element.

How? Can use struct / class.

struct: allows you to define your own datatype, made from existing types.

Example:
```
struct point {
    double x;
    double y;
};
```

usage:

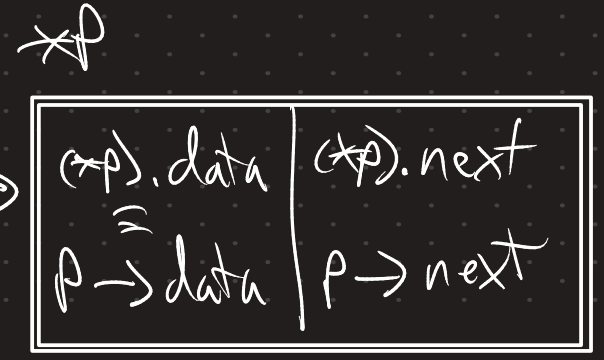
```
point p; // declare a point
p.x = 1.0;
p.y = 2.0;
```

What about our list situation?

```
struct node {
    int data;
    node* next;
};
```

More pictures:

```
node n;
```

```
node* p = new node;
```

(much more common)